

Mini SOC Lab using Wazuh SIEM

Project Objective: To build a Blue Team Mini SOC Lab using Ubuntu Wazuh Server and a Windows 11 VM for detecting brute-force attacks, threat hunting, and MITRE ATT&CK mapping.

Lab Architecture

Windows 11 VM (Wazuh Agent) → Wazuh Server (Ubuntu VM) → Threat Hunting Dashboard

Tools Used

- VirtualBox
- Ubuntu Server
- Wazuh SIEM
- Windows 11 VM
- PowerShell
- Threat Hunting Dashboard

Task Performed: Brute Force Detection

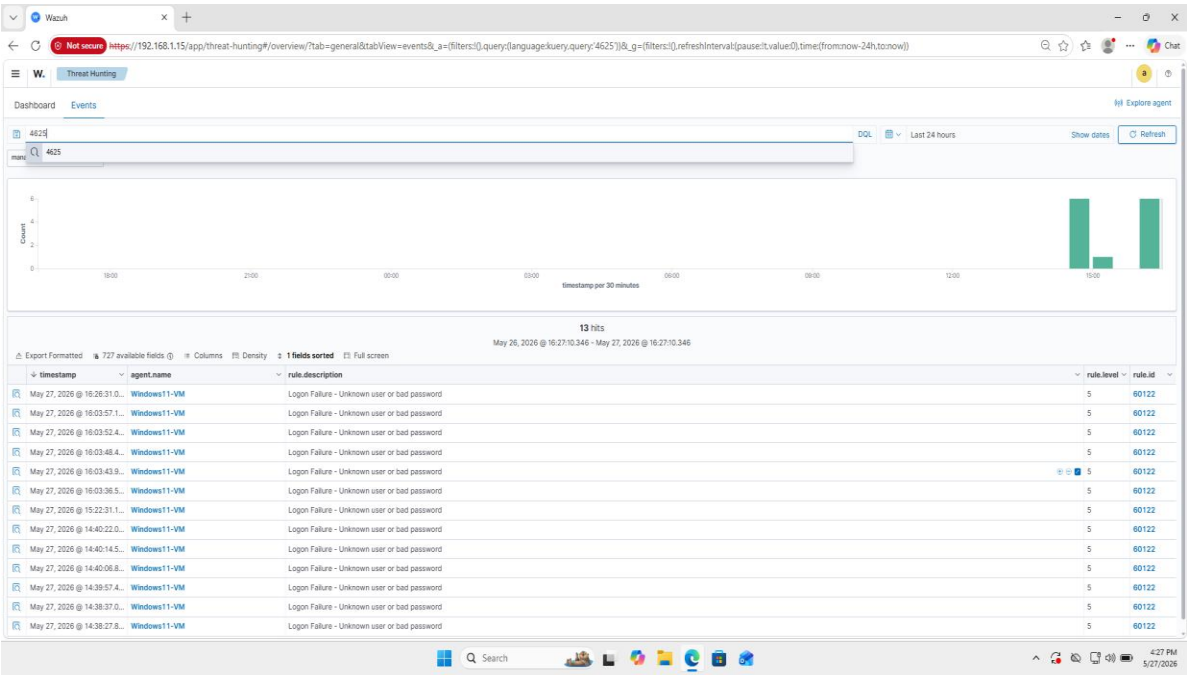
The Windows 11 VM was locked using Win + L and multiple incorrect passwords were entered intentionally to simulate a brute-force attack.

Detection Details

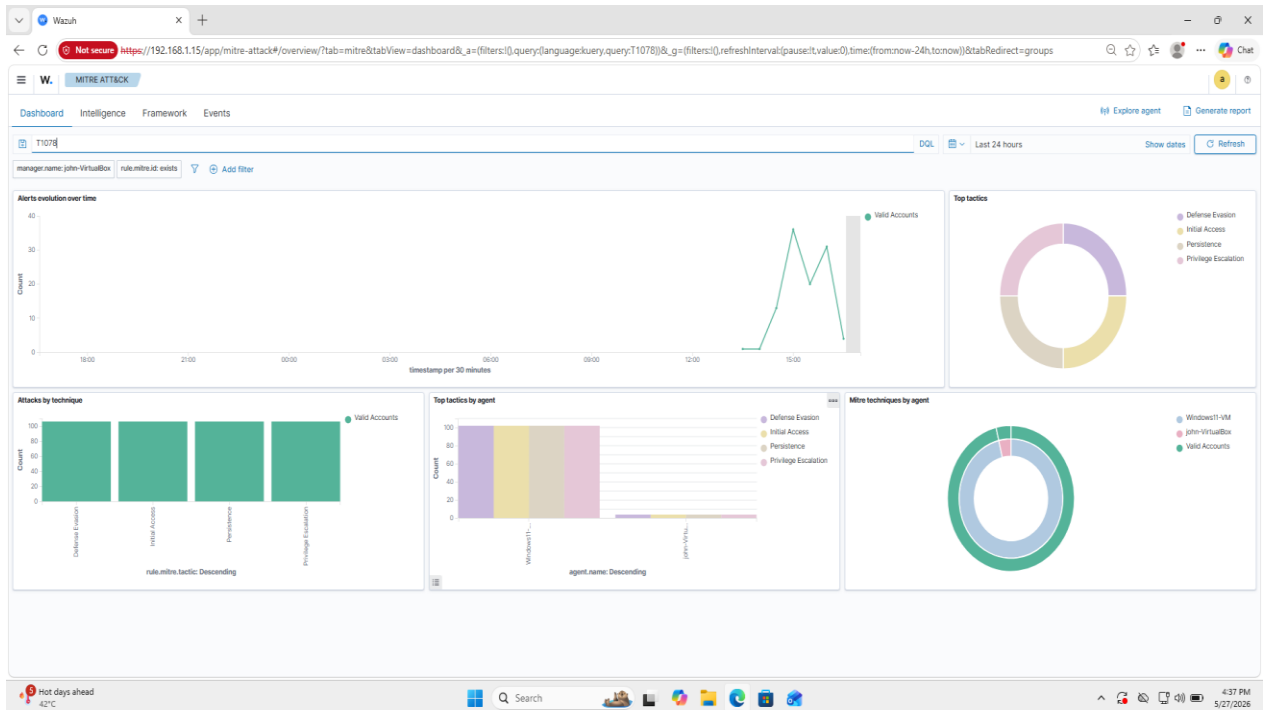
Field	Value
Detection Name	Windows Brute Force Detection
Event ID	4625
Log Source	Windows Security Event Logs
MITRE ATT&CK	T1110 – Brute Force
Platform	Windows 11 VM

Threat Hunting Results

Wazuh SIEM successfully detected multiple failed login attempts and displayed them in the Threat Hunting dashboard.



Threat Hunting dashboard showing failed login attempts.



MITRE ATT&CK mapping dashboard showing T1078 Valid Accounts.

Conclusion

The Mini SOC Lab was successfully configured using Wazuh SIEM. Threat Hunting and MITRE ATT&CK mapping were implemented successfully, allowing detection of brute-force login attempts and account-related attack activity.